

VEHICLE SERVICE MANAGEMENT SYSTEM USING CRUD OPERATION

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1. Introduction

The Vehicle Service Management System is a web-based application developed to manage vehicle service records digitally. The system allows the user to add, view, update and delete vehicle service information.

The application stores service records in a MySQL database and uses Node.js and Express.js for backend operations. HTML, CSS and JavaScript are used to create the user interface.

The screenshot displays the 'Vehicle Service Management System' web application. At the top, there are four summary cards: 'Total Services' (0), 'Pending' (0), 'In Progress' (0), and 'Completed' (0). Below these is the 'Add Service Record' form, which includes input fields for Name, Phone Number, Email ID, Vehicle Number, Vehicle Brand, Vehicle Model, Service Type, Service Date, and Service Status. A '+ Add Service' button is located at the bottom of the form. Below the form is a 'Service Records' table with columns: NO., NAME, PHONE, EMAIL, VEHICLE NO., BRAND, MODEL, SERVICE TYPE, DATE, STATUS, and ACTIONS. The footer of the application shows the text 'Vehicle Service Management System • CRUD Application • by Harish'.

2. Technologies Used

The following technologies are used in this project:

Technology	Purpose
HTML	Creates the structure of the webpage
CSS	Designs and styles the application
JavaScript	Handles frontend functionality
Node.js	Runs the backend server
Express.js	Creates the REST API
MySQL	Stores service records
MySQL Workbench	Manages the database
VS Code	Used for development

3. Project Structure

The project contains the following files:

crud_project/

```
|
| └─ public/
|   └─ index.html
|   └─ style.css
|   └─ script.js
|
| └─ db.js
| └─ server.js
| └─ schema.sql
| └─ package.json
└─ package-lock.json
```

4. Database Connection

db.js

```
const mysql = require("mysql2");
const db = mysql.createPool({
  host: "localhost",
  user: "root",
  password: "root",
  database: "vehicle_service"
});
module.exports = db.promise();
```

SQL

schema.sql

```
CREATE DATABASE IF NOT EXISTS vehicle_service;
```

```
USE vehicle_service;
```

```
CREATE TABLE IF NOT EXISTS service_records (  
  id INT AUTO_INCREMENT PRIMARY KEY,
```

```
  customer_name VARCHAR(100) NOT NULL,
```

```
  phone_number VARCHAR(20) NOT NULL,
```

```
  email_id VARCHAR(150) NOT NULL,
```

```
  vehicle_number VARCHAR(30) NOT NULL,
```

```
  vehicle_brand VARCHAR(100) NOT NULL,
```

```
  vehicle_model VARCHAR(100) NOT NULL,
```

```
  service_type VARCHAR(100) NOT NULL,
```

```
  service_date DATE NOT NULL,
```

```
  service_status VARCHAR(50) NOT NULL
```

```
);
```

```
SELECT * FROM service_records;
```

Output	Result #4								
id	customer_name	phone_number	email_id	vehicle_number	vehicle_brand	vehicle_model	service_type	service_date	service_status
6	harish	1234567890	harish@gmail.com	KA51EY9085	Royal Enfield	Continental GT 650	General Service	09/29/2026	In Progress
7	jawad	1234567890	jawad@gmail.com	KA01JD1234	Royal Enfield	Interceptor 650	General Service	09/30/2026	Pending
8	sai tharun	1234567890	sai tharun@gmail.com	KA51ST1234	Bajaj	Pulsar 150	Oil Change	09/28/2026	Pending

5. Backend Server

Server.js

```
const express = require("express");
```

```

const path = require("path");
const db = require("./db");
const app = express();
const PORT = 3000;
app.use(express.json());
app.use(
  express.urlencoded({
    extended: true
  })
);
app.use(
  express.static(
    path.join(__dirname, "public")
  )
);

```

6. CRUD OPERATION

The project implements all four basic database operations.

Operation	HTTP Method	Purpose
Create	POST	Add service record
Read	GET	Display service records
Update	PUT	Edit service record
Delete	DELETE	Delete service record

CREATE → POST

READ → GET

UPDATE → PUT

DELETE → DELETE

7.CRUD OPERATION

i. Server.js

create

```
app.post("/api/services", async (req, res) => {
  try {
    const {
      customer_name,
      phone_number,
      email_id,
      vehicle_number,
      vehicle_brand,
      vehicle_model,
      service_type,
      service_date,
      service_status
    } = req.body;
    // VALIDATION
    if (
      !customer_name ||
      !phone_number ||
      !email_id ||
      !vehicle_number ||
      !vehicle_brand ||
      !vehicle_model ||

      !service_type ||
      !service_date ||
      !service_status
    ) {
      return res.status(400).json({
        error:
          "Please fill all fields"
      });
    }
    // PHONE VALIDATION
    if (!/^[0-9]{10}$/.test(phone_number)) {
      return res.status(400).json({
        error:
          "Phone number must contain 10 digits"
      });
    }
    // INSERT
    const [result] = await db.query(
      INSERT INTO service_records
      (
        customer_name,
        phone_number,
```

```

        email_id,
        vehicle_number,
        vehicle_brand,
        vehicle_model,
        service_type,
        service_date,
        service_status
    )
VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?)
`,
    [
        customer_name,
        phone_number,
        email_id,
        vehicle_number,
        vehicle_brand,
        vehicle_model,
        service_type,
        service_date,
        service_status
    ]
);
res.status(201).json({
    message:
        "Service record added successfully",
    id:
        result.insertId
});
} catch (error) {
    console.error("POST ERROR:", error);
    res.status(500).json({
        error:
            "Failed to add service record"
    });
}
});

```

Vehicle Service Management System

Total Services: 3 | Pending: 3 | In Progress: 0 | Completed: 0

Add Service Record
Enter customer and vehicle service details

Name: shankar | Phone Number: 1234567890 | Email ID: shankar@gmail.com
 Vehicle Number: ka51ta1234 | Vehicle Brand: KTM | Vehicle Model: RC 390
 Service Type: Engine Service | Service Date: 30-09-2026 | Service Status: Pending

Service Records
Manage all vehicle service records

ID	NAME	PHONE	EMAIL	VEHICLE NO.	BRAND	MODEL	SERVICE TYPE	DATE	STATUS	ACTIONS
1	harish	1234567890	harish@gmail.com	KA31E790E3	Royal Enfield	Continental GT 650	General Service	2026-09-29	Pending	Edit Delete
2	javed	1234567890	javed@gmail.com	KA01L01234	Royal Enfield	Interceptor 650	General Service	2026-09-29	Pending	Edit Delete
3	sai tharan	1234567890	satharan@gmail.com	KA31S1234	Bajaj	Pulsar 150	Oil Change	2026-09-27	Pending	Edit Delete

ii. read

```
app.get("/api/services", async (req, res) => {
  try {
    const [rows] = await db.query(`
      SELECT *
      FROM service_records
      ORDER BY id ASC
    `);
    res.json(rows);
  } catch (error) {
    console.error("GET ERROR:", error);
    res.status(500).json({
      error:
        "Failed to load service records"
    });
  }
});
```

Vehicle Service Management System

Total Services
4

Pending
4

In Progress
0

Completed
0

Add Service Record

Enter customer and vehicle service details

Name

Phone Number

Email ID

Vehicle Number

Vehicle Brand

Vehicle Model

Service Type

Service Date

Service Status

[+ Add Service](#)

Service Records

Manage all vehicle service records

SL	NAME	PHONE	EMAIL	VEHICLE SL	BRAND	MODEL	SERVICE TYPE	DATE	STATUS	ACTIONS
1	harish	1234567890	harish@gmail.com	KA512Y9085	Royal Enfield	Continental GT 650	General Service	2026-09-29	Pending	Edit Delete
2	javed	1234567890	javed@gmail.com	KA01B1234	Royal Enfield	Interceptor 650	General Service	2026-09-29	Pending	Edit Delete
3	sai tharan	1234567890	satharun@gmail.com	KA512T1234	Bajaj	Pulsar 150	Oil Change	2026-09-27	Pending	Edit Delete
4	chankar	1234567890	chankar@gmail.com	KA515A1234	KTM	RC 390	Engine Service	2026-09-29	Pending	Edit Delete

iii. update

```
app.put("/api/services/:id", async (req, res) => {
  try {
    const { id } = req.params;
    const {
      customer_name,
      phone_number,
      email_id,
      vehicle_number,
      vehicle_brand,
      vehicle_model,
      service_type,
      service_date,
      service_status
    } = req.body;
    // VALIDATION
    if (
      !customer_name ||
      !phone_number ||
      !email_id ||
      !vehicle_number ||
      !vehicle_brand ||
      !vehicle_model ||
      !service_type ||
      !service_date ||
      !service_status
    ) {
      return res.status(400).json({
```

```

        error:
            "Please fill all fields"
    });
}
if (!/^[0-9]{10}$/.test(phone_number)) {
    return res.status(400).json({
        error:
            "Phone number must contain 10 digits"
    });
}
// UPDATE
const [result] = await db.query(
    `
    UPDATE service_records
    SET
        customer_name = ?,
        phone_number = ?,
        email_id = ?,
        vehicle_number = ?,
        vehicle_brand = ?,
        vehicle_model = ?,
        service_type = ?,
        service_date = ?,
        service_status = ?
    WHERE id = ?
    `,
    [
        customer_name,
        phone_number,
        email_id,
        vehicle_number,
        vehicle_brand,
        vehicle_model,
        service_type,
        service_date,
        service_status,
        id
    ]
);
if (result.affectedRows === 0) {
    return res.status(404).json({
        error:
            "Service record not found"
    });
}
res.json({
    message:
        "Service record updated successfully"
});
} catch (error) {

```

```

console.error("PUT ERROR:", error);
res.status(500).json({
  error:
    "Failed to update service record"
});
}
});

```

! for better understanding see this status bar
before update

Vehicle Service Management System

Total Services
4

Pending
4

In Progress
0

Completed
0

Update Service Record

Enter customer and vehicle service details

Name

Phone Number

Email ID

Vehicle Number

Vehicle Brand

Vehicle Model

Service Type

Service Date

Service Status

Service Records

Manage all vehicle service records

NO.	NAME	PHONE	EMAIL	VEHICLE NO.	BRAND	MODEL	SERVICE TYPE	DATE	STATUS	ACTIONS
1	harish	1234567890	harish@gmail.com	KA51EY9085	Royal Enfield	Continental GT 650	General Service	2026-09-29	Pending	Edit Delete
2	jamad	1234567890	jamad@gmail.com	KA01JD1234	Royal Enfield	Interceptor 650	General Service	2026-09-29	Pending	Edit Delete
3	sai tharan	1234567890	saiharun@gmail.com	KA51ST1234	Bajaj	Pulsar 150	Oil Change	2026-09-27	Pending	Edit Delete
4	shankar	1234567890	shankar@gmail.com	KA51SA1234	KTM	RC 390	Engine Service	2026-09-29	Pending	Edit Delete

After update

Vehicle Service Management System

Total Services
4

Pending
3

In Progress
1

Completed
0

Add Service Record

Enter customer and vehicle service details

Name
Enter Customer Name

Phone Number
Enter 10-digit Phone Number

Email ID
Enter Email ID

Vehicle Number
KA01AB1234

Vehicle Brand
Select Vehicle Brand

Vehicle Model
Select Vehicle Model

Service Type
Select Service Type

Service Date
dd-mm-yyyy

Service Status
Select Status

+ Add Service

Service Records

Manage all vehicle service records

NO.	NAME	PHONE	EMAIL	VEHICLE NO.	BRAND	MODEL	SERVICE TYPE	DATE	STATUS	ACTIONS
1	harish	1234567890	harish@gmail.com	KA51EY9005	Royal Enfield	Continental GT 650	General Service	2026-09-28	In Progress	Edit Delete
2	jawad	1234567890	jawad@gmail.com	KA01JD1234	Royal Enfield	Interceptor 650	General Service	2026-09-29	Pending	Edit Delete
3	sai tharun	1234567890	saiharun@gmail.com	KA51ST1234	Bajaj	Pulsar 150	Oil Change	2026-09-27	Pending	Edit Delete
4	shankar	1234567890	shankar@gmail.com	KA51SA1234	KTM	RC 390	Engine Service	2026-09-29	Pending	Edit Delete

iv. delete

```
app.delete("/api/services/:id", async (req, res) => {
  try {
    const { id } = req.params;
    const [result] = await db.query(
      `DELETE FROM service_records
      WHERE id = ?`,
      [id]
    );
    if (result.affectedRows === 0) {
      return res.status(404).json({
        error:
          "Service record not found"
      });
    }
  }
  res.json({
    message:
```

```

        "Service record deleted successfully"
    });
} catch (error) {
    console.error("DELETE ERROR:", error);
    res.status(500).json({
        error:
            "Failed to delete service record"
    });
}
});

```

before delete

Vehicle Service Management System

Total Services
4

Pending
3

In Progress
1

Completed
0

Add Service Record

Enter customer and vehicle service details

Name

Phone Number

Email ID

Vehicle Number

Vehicle Brand

Select Vehicle Brand

Vehicle Model

Select Vehicle Model

Service Type

Select Service Type

Service Date

Service Status

Select Status

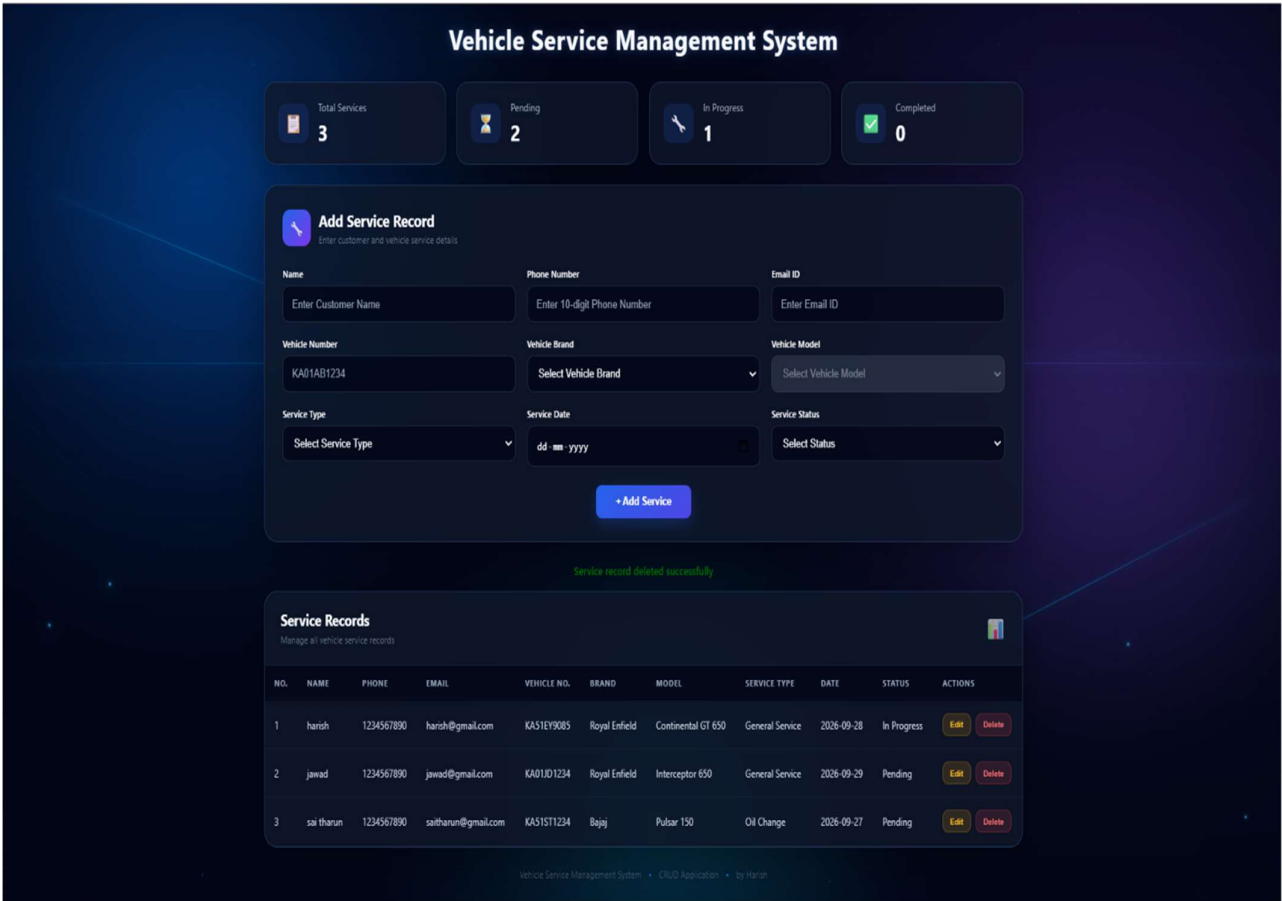
[+ Add Service](#)

Service Records

Manage all vehicle service records

NO.	NAME	PHONE	EMAIL	VEHICLE NO.	BRAND	MODEL	SERVICE TYPE	DATE	STATUS	ACTIONS
1	harish	1234567890	harish@gmail.com	KA51EY9085	Royal Enfield	Continental GT 650	General Service	2026-09-28	In Progress	Edit Delete
2	jawad	1234567890	jawad@gmail.com	KA01UD1234	Royal Enfield	Interceptor 650	General Service	2026-09-29	Pending	Edit Delete
3	sai thanun	1234567890	saithanun@gmail.com	KA51ST1234	Bajaj	Pulsar 150	Oil Change	2026-09-27	Pending	Edit Delete
4	shankar	1234567890	shankar@gmail.com	KA51SA1234	KTm	RC 390	Engine Service	2026-09-29	Pending	Edit Delete

after delete



8.Package Configuration –

package.json

The package.json file contains the basic configuration of the Node.js project. It defines the project name, version, start command, and required dependencies.

```
{  
  "name": "vehicle-service-management",  
  "version": "1.0.0",  
  "description": "Vehicle Service Management System",  
  "main": "server.js",  
  "scripts": {  
    "start": "node server.js"  
  },  
  "dependencies": {  
    "express": "^5.1.0",  
    "mysql2": "^3.14.3"  
  }  
}
```

9. Running the Project

The project is started using:

npm start

The server runs at:

<http://localhost:3000>

```
PS C:\Users\JEEVAN HARISHA\OneDrive\Desktop\crud_project> npm start
> vehicle-service-management@1.0.0 start
> node server.js

ready to go
  Server running at http://localhost:3000
|
```

10. Final Output

The completed **Vehicle Service Management System** provides a simple interface for managing vehicle service records. The application connects the frontend with a Node.js/Express backend and MySQL database and supports complete CRUD functionality.

The screenshot displays the 'Vehicle Service Management System' web application. At the top, there are four summary cards: 'Total Services' (4), 'Pending' (3), 'In Progress' (1), and 'Completed' (0). Below these is the 'Add Service Record' form, which includes fields for Name, Phone Number, Email ID, Vehicle Number, Vehicle Brand, Vehicle Model, Service Type, Service Date, and Service Status. A '+Add Service' button is at the bottom of the form. The bottom section, titled 'Service Records', contains a table with 4 columns: ID, NAME, PHONE, EMAIL, VEHICLE NO., BRAND, MODEL, SERVICE TYPE, DATE, STATUS, and ACTIONS. The table lists four service records with their respective details and 'Edit' and 'Delete' buttons for each.

ID	NAME	PHONE	EMAIL	VEHICLE NO.	BRAND	MODEL	SERVICE TYPE	DATE	STATUS	ACTIONS
1	harish	1234567890	harish@gmail.com	KAS1EY9085	Royal Enfield	Continental GT 650	General Service	2026-09-28	In Progress	Edit Delete
2	javed	1234567890	javed@gmail.com	KAB1UD1234	Royal Enfield	Interceptor 650	General Service	2026-09-29	Pending	Edit Delete
3	sai tharan	1234567890	satharan@gmail.com	KAS1ST1234	Bajaj	Pulsar 150	Oil Change	2026-09-27	Pending	Edit Delete
4	shankar	1234567890	shankar@gmail.com	KAS1SA1234	KTM	RC 390	Engine Service	2026-09-29	Pending	Edit Delete